

To XOR a Stone with Six Birds: Closure Diagnostics for Emergent Bits, Gates, and Booleanity

Ioannis Tsiokos 

v1: March 9, 2026 · v3: October 3, 2026

Abstract

When does a physical or stochastic system deserve to be described in terms of bits and logic gates? We treat logic not as a starting point but as something a coarse-grained description has to earn. Working in the Six Birds emergence calculus, we call a description a *logic layer* when it is an audited closure of packaged predicates under induced dynamics: its binary labels persist, the underlying dynamics descends to update rules on those labels, and what the description gets wrong or leaves out is measured explicitly. We turn this definition into a set of diagnostics (predicate stability, route mismatch, closure defect, channel error and entropy, unretained input information, and entropy-production audits) and apply them to finite Markov laboratories in which NOT, AND, and reversible XOR (CNOT) gates run under controlled noise, metastability, and coarse-graining. Four results follow. First, in a laboratory with leaky parity sectors, the parity bit is an exactly closed coarse variable, with route mismatch zero at every leakage rate and horizon, whereas the other balanced binary partitions and the AND predicate have strictly positive mismatch at every tested leakage rate and horizon; closed-form expressions give the size of the gap. Second, the three gates fail in different ways: NOT and AND lose closure when their stored inputs drift, whereas a noiseless one-step CNOT is a perfect input–output channel even though its output bit, read alone, is maximally unclosed. Third, reading the same CNOT laboratory through its XOR output instead of its full register raises the input information left unaccounted for by the output from $h_2(0.02) \approx 0.141$ bits to $1 + h_2(0.02)$ bits, a factor of 8.07 that rational bounds certify to exceed eight, and raises route mismatch from 0 to 0.96. Fourth, a simple unsupervised pipeline recovers the intended stable bits from transition structure alone and reconstructs a NOT gate in declared label coordinates. The predicate and quotient facts on which the framework rests are formalized in Lean 4, and the main experimental quantities are re-derived independently in closed form, with their probabilities and route mismatches checked in exact rational arithmetic. The result is a falsifiable, layer-relative criterion for when a system has logic, and for which logic it has.

Keywords: emergent logic; coarse-graining; lumpability; closure diagnostics; parity and XOR; reversible computation; information loss.

Contribution summary. We define a logic layer as an audited closure of packaged predicates, give diagnostics that test each part of that definition, and show in controlled finite laboratories that parity is an exactly closed coarse variable, that reversible and erased views of one gate are different logical objects, and that stable bits and a NOT gate can be recovered from transition structure without being declared in advance.

1 Introduction

Most accounts of physical computation start from bits, registers, and truth tables, and then ask how a substrate implements them and at what thermodynamic cost [1–4]. This paper asks an earlier question: when does a coarse-grained description of a system deserve to be called logical at all? Any system can be given labels that look like 0 and 1. The labels may still fail to persist, the dynamics may not act on them as a well-defined rule, and the description may silently discard information that the system still carries. In such cases logic has been projected onto the system, not found in it.

Our answer is operational. We work in the Six Birds emergence calculus [5], which describes how new layers of description arise from six primitives: operator rewrite (P1), gating and constraints (P2), protocol holonomy (P3), sectors and invariants (P4), packaging (P5), and accounting and audit (P6). Logic is not a seventh primitive in this picture. It is something these six can jointly produce. A system has a *logic layer* when packaged binary predicates persist, when the dynamics induces well-defined update rules on them, and when the reliability and the losses of that closure are measured. In one sentence: a logic layer is an audited closure of packaged predicates under induced dynamics.

The definition has two sides. On the semantic side, a proposition is a set of microstates that the layer can express, and the connectives are set operations on those sets. On the operational side, a bit is an equivalence class of microstates, and a gate is an update that the micro-dynamics induces on such classes. The first side says what the layer’s statements mean; the second says whether the substrate can execute them. Replacing “how is logic implemented?” with “when has logic emerged strongly enough to close?” connects the question to classical work on rough sets and partition logic [6, 7] and to the theory of macrostates and coarse variables [8].

The title emphasizes XOR for a concrete reason. Variables aligned with a system’s sector or invariant structure are often especially good coarse variables [8–10], and parity is the simplest such variable. Parity is not metaphysically special. It is a clean test case for the paper’s broader claim that some logical distinctions are easier for a substrate to stabilize than others. The useful question is therefore not whether a substrate can be made to emulate a given truth table in principle, but which distinctions it supports natively, how robustly, and at what accounting cost.

To make the definition testable we introduce a small suite of diagnostics: predicate stability, route mismatch, closure defect, gate error, conditional entropy, unretained input information, and entropy production. Together they ask whether a packaged bit persists long enough to matter, whether microstates that share a label also share a coarse future, whether a gate is a closed operator or an artifact of how it was evaluated, and whether information that seems to disappear has been destroyed or merely left out of the description. Formally, the descent of dynamics to coarse variables is the lumpability problem for Markov chains [9, 11–13]; the comparison of retained and erased descriptions connects to the literature on irreversibility and reversible computation [1, 2, 4]. Within the Six Birds program, the paper builds on the foundations in [5] and on related closure analyses in [14, 15].

We apply the diagnostics to finite Markov laboratories in which NOT, AND, and reversible XOR (CNOT) gates run with tunable noise, metastability, and coarse-graining, and ask four questions. Which binary coarse variables close most robustly? How do familiar gates behave as noise, staging barriers, and evaluation horizons vary? What changes when a reversible gate is read through a simpler but lossy output? And can stable bits and gates be recovered from transition structure without being declared in advance?

The answers are as follows. In a laboratory built around leaky parity sectors, the parity bit has zero route mismatch at every leakage rate and horizon, whereas every other balanced binary partition and the AND predicate have strictly positive mismatch at every tested leakage rate and

horizon, with closed-form expressions for each (Section 4). NOT and AND lose closure when their stored inputs drift over long horizons, whereas CNOT separates two properties that usually travel together: a noiseless one-step CNOT is an exact input–output channel, yet its output bit, read alone, has maximal route mismatch (Section 5). Reading one CNOT laboratory through its XOR output rather than its full register multiplies the unretained input information by 8.07 and raises route mismatch from 0 to 0.96, so the reversible gate and its erased view are different logical objects rather than one object described at two resolutions (Section 6). Finally, a spectral and information-theoretic pipeline recovers the intended stable bits and a NOT gate directly from the transition kernel (Section 7).

These results have different epistemic status, and we keep them apart. The predicate and quotient facts behind the framework, such as the closure of definable predicates under the Boolean connectives and the descent of updates that respect a partition, are proved in Lean 4. The main laboratory quantities, including the parity comparison, the reversible-versus-erased comparison, and the gate channels in every condition of the phase diagram, are re-derived independently in closed form. Their probabilities and route mismatches are checked in exact rational arithmetic, using the recorded decimal values of the barrier-dependent memory noise in the phase grid; Shannon quantities are checked numerically, and the eightfold loss inequality is certified with rational logarithm bounds (Appendix B). Everything else comes from floating-point computation and is reported as such.

Contributions. The paper makes three contributions. First, it gives a layer-relative definition of logic within the Six Birds framework: a logic layer is an audited closure of packaged predicates under induced dynamics. Second, it provides diagnostics that test each clause of that definition and separate closed logic layers from symbolic descriptions that merely fit. Third, it demonstrates the framework in controlled finite laboratories, establishing exact parity closure, a separation between reversible and erased views of one gate, and unsupervised recovery of packaged bits and a NOT gate.

Scope. The paper is neither a survey of physical computation nor a claim that Boolean logic is the only possible outcome of coarse-graining. In the set-based laboratories studied here, the events a lens can express always form a Boolean algebra, so complement is never unavailable as a statement. What a layer can lack is the ability to keep its bits stable and to execute operations on them, and these are what the diagnostics measure. Showing that a layer has a genuinely non-Boolean event structure would require restricted operations or richer semantics, which we do not develop here. The laboratories are designed for diagnostic clarity rather than physical realism, and Definition 1 is an operational criterion whose tolerances are left to the application.

Section 2 defines logic layers through lenses, packaging, and induced dynamics. Section 3 introduces the diagnostics and the laboratories. Sections 4–7 present the four results, and Section 8 discusses what they imply about logic as a property of layers. Appendix A describes reproduction, Appendix B describes the formal and exact support, and Appendix C records the version history.

2 Logic Layers, Packaging, and Closure

We do not rederive the full Six Birds formalism here [5]. Four ingredients are enough to talk about logic: a lens that fixes what the layer can observe, a packaging step that turns observations into reusable binary carriers, an induced dynamics that may or may not close on those carriers, and an audit that measures persistence, mismatch, and loss. In Six Birds terms, packaging (P5) supplies the carriers, constraints (P2) give them meaning, sectors and metastability (P4) keep them stable,

operator rewrite (P1) governs which effective operators can be introduced, protocol holonomy (P3) shows up as dependence on the order of evaluation, and accounting (P6) keeps the description honest. This section turns these ingredients into a definition of a *logic layer*.

2.1 Lenses, definability, and packaged predicates

Let Z be a finite set of microstates and let

$$f : Z \rightarrow X$$

be a lens, or coarse-graining map, into a set of labels X . The fibers $f^{-1}(x)$ are the groups of microstates that the layer does not tell apart. A binary predicate $h : Z \rightarrow \{0, 1\}$ is *definable* at the lens if it factors through it, that is, if

$$h = \tilde{h} \circ f$$

for some macro-predicate $\tilde{h} : X \rightarrow \{0, 1\}$. This holds exactly when h is constant on every fiber of f . The equivalence needs no assumption that f is onto; when it is not, $\tilde{h}$ is determined on the image of f and arbitrary elsewhere. We write Σ_f for the collection of definable predicates, or equivalently of definable events $A \subseteq Z$ (unions of fibers). Σ_f is the expressive content of the layer: a distinction that is not definable at f is not available as a statement at that layer [6–8].

Two consequences are immediate, and both are proved in our Lean formalization (Appendix B). First, Σ_f contains the always-true and always-false predicates and is closed under conjunction, disjunction, negation, and exclusive or; it is a Boolean algebra. Second, definability is inherited along any update that descends to the labels: if $F : Z \rightarrow Z$ satisfies $f \circ F = g \circ f$ for some $g : X \rightarrow X$, then $h \circ F$ is definable whenever h is.

Packaging is the step that turns a definable distinction into a reusable carrier. In the binary case a package is a map $\Pi : Z \rightarrow \{0, 1\}$ that assigns many microstates to the same label because the layer treats them as equivalent. For a definable event A , the packaged predicate is its indicator $\Pi_A(z) = \mathbf{1}[z \in A]$. Truth values are thus compressed labels of microstates rather than primitive objects. They are useful only if they persist over the timescale on which the layer operates, which is where sectors and metastability (P4) enter [8, 9, 16].

2.2 Two complementary views of logic

In the *feasibility view*, propositions are definable sets of admissible microstates and connectives are set operations on them:

$$A \wedge B = A \cap B, \quad A \vee B = A \cup B, \quad \neg A = Z \setminus A, \quad A \oplus B = (A \setminus B) \cup (B \setminus A).$$

Logic in this view is not an extra layer placed on top of the dynamics. It is the algebra of distinctions that a lens can express. Constraints (P2) appear as admissibility: a proposition restricts which states are allowed, and a contradiction is the empty set. Because Σ_f is a Boolean algebra, every connective of definable events is again definable. In particular, the complement of a definable event is always available *as a statement*. Whether the substrate can *execute* negation, that is, whether some dynamics maps each state carrying a bit to a state carrying its complement, is a separate operational question.

In the *operator view*, a packaged bit matters only if the dynamics induces updates on it. For a deterministic micro-update $F : Z \rightarrow Z$, an induced macro-update $g : X \rightarrow X$ exists if and only if F respects the fibers, meaning that $f(z) = f(z')$ implies $f(F(z)) = f(F(z'))$. When it exists, g is unique on the image of f , and induced updates compose: the update induced by $F_2 \circ F_1$ is the

composite of the updates induced by F_1 and F_2 . For stochastic dynamics given by a Markov kernel P on Z , the corresponding requirement is lumpability [9, 11–13]. Write

$$R_\tau(i, x) = \sum_{j: f(j)=x} (P^\tau)_{ij}$$

for the probability that microstate i reaches label x after τ steps. The lens is lumpable at horizon τ when $R_\tau(i, \cdot)$ is the same for all i in each fiber; the common rows then form a macro-kernel $K_{\tau, f}$ on the image $f(Z)$, and evolving and then coarse-graining agrees with coarse-graining and then evolving:

$$Q_f(\mu P^\tau) = Q_f(\mu) K_{\tau, f} \quad \text{for every distribution } \mu \text{ on } Z,$$

where Q_f pushes a distribution forward through the lens. Section 3 measures how far a lens is from this condition.

It is also useful to have a map that goes back from labels to microstates. Fix for each nonempty fiber a probability vector supported on that fiber (its *prototype*; we use the uniform one) and let U_f send a macro-distribution supported on $f(Z)$ to the corresponding mixture of prototypes. Then $Q_f U_f$ is the identity on macro-distributions supported on the image of f , and the packaging map $T = U_f Q_f$ is idempotent: packaging twice is the same as packaging once. The dynamical map

$$E_{\tau, f}(\mu) = U_f(Q_f(\mu P^\tau))$$

evolves the micro-distribution, reduces it through the lens, and lifts the result back to a packaged micro-distribution. Unlike T , the map $E_{\tau, f}$ is in general not idempotent.

The two views need each other. Feasibility without operator closure gives semantics that the substrate cannot execute. Operator closure without feasibility semantics gives a convenient signal-processing description, but not yet a logic. A logic layer appears when the same packaged distinctions support both.

Definition 1 (Logic layer). Fix a microstate space Z , a lens $f : Z \rightarrow X$, and an evaluation window τ . A *logic layer* at (f, τ) is a nontrivial family $\mathcal{B} \subseteq \Sigma_f$ of binary packaged predicates, together with induced macro-updates, such that:

1. **Stability.** Each $b \in \mathcal{B}$ is metastable on the timescale τ .
2. **Closure.** The relevant micro-updates descend to macro-updates on the packaged variables with small closure defect.
3. **Feasibility semantics.** Each $b \in \mathcal{B}$ is the indicator of a definable feasible set, and the connectives the layer uses are definable and stable at (f, τ) .
4. **Auditability.** Persistence, agreement between evaluation routes, and information loss are measured by explicit diagnostics.

When these conditions fail, a symbolic description may still be convenient, but it is not a closed logic layer in the sense of this paper.

Definition 1 is an operational criterion, not a theorem. It does not fix numerical thresholds for “metastable” or “small”; those depend on the application, and the experiments below report the diagnostics themselves rather than a single pass/fail verdict. In the set-based laboratories of this paper, clause 3 adds little beyond clause 1. Definability of the connectives is automatic, and their stability is controlled by that of their inputs: in the stability measure S_τ of Section 3, a predicate

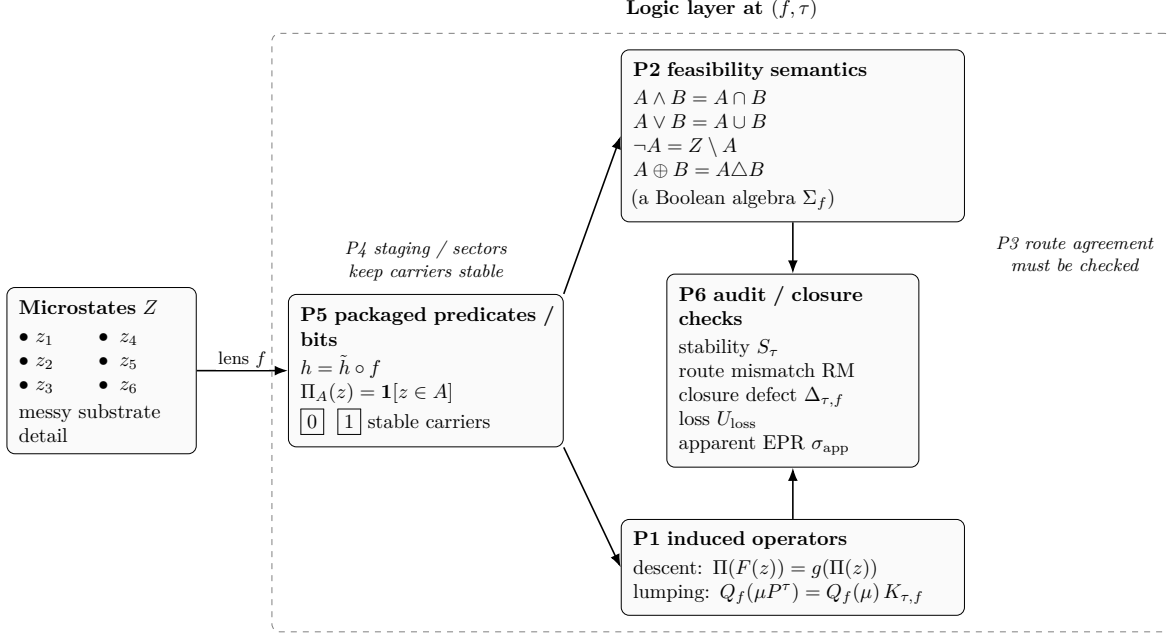


Figure 1. A logic layer is an audited closure of packaged predicates. A lens f and a packaging step compress microstates into binary carriers, kept stable by sector structure or metastability. The carriers support two views at once: feasibility semantics on definable sets, which form a Boolean algebra Σ_f , and macro-operators induced by the dynamics, either by exact descent of a deterministic update or by lumping a Markov kernel. The layer counts as logical only when the resulting closure survives explicit audit through stability, route mismatch, closure defect, unretained input information, and entropy-production diagnostics.

and its complement are equally stable, and a connective of two predicates b and c can change value only if b or c does, so

$$S_\tau(b \star c) \geq 1 - (1 - S_\tau(b)) - (1 - S_\tau(c))$$

for any binary connective $\star$. The clause carries more weight in settings where admissible states or executable operations are restricted.

Figure 1 summarizes the structure: microstates enter the layer only through packaging, and the resulting predicates count as logical only when their feasibility semantics and their induced operators both survive audit.

2.3 Why the criterion is layer-relative

Definition 1 asks for more than the ability to label states by zero and one. A candidate bit must last long enough to matter, must support induced operators, and must survive audit. This is why the six primitives work together rather than overlap: P5 packages the carrier, P4 keeps it from dissolving, P2 makes it meaningful, P1 determines whether an effective operator vocabulary can be stabilized, P3 detects dependence on evaluation order and hidden protocol state, and P6 records what it costs to maintain the arrangement.

The criterion is also deliberately broader than classical Booleanity. In the laboratories studied here the expressible events always form a Boolean algebra, so what varies from layer to layer is not whether complements exist as statements but which operations the dynamics carries out reliably, at which horizon, and on which carrier. A layer may support a stable parity bit but no reliable conjunction, or may support a reversible operator only on a larger carrier than the one being read. We therefore speak of a *logic layer* rather than a *Boolean layer*. Genuinely non-Boolean

event structures, of the kind studied in quantum logic and partition logic [7, 17, 18], would require restricting the admissible operations or enriching the semantics, which the present laboratories do not do.

Section 3 turns the four clauses into measurements: predicate stability for the first clause, route mismatch and closure defect for the second, feasible-set and gate semantics for the third, and unretained input information and entropy production for the fourth.

3 Diagnostics and Methods

This section turns the four clauses of Definition 1 into measurable quantities and describes the laboratories in which we measure them. Following the broader Six Birds practice of pairing every closure claim with an explicit audit [14, 15], all numbers in Sections 4–7 are read from a frozen, hash-checked result bundle that the repository regenerates end to end (Appendix A).

3.1 Stability and closure diagnostics

Predicate stability. For a binary predicate $b : Z \rightarrow \{0, 1\}$, a Markov kernel P on Z , a horizon τ , and an initial distribution μ , the stability

$$S_\tau(b; \mu) = \sum_{i,j} \mu_i (P^\tau)_{ij} \mathbf{1}[b(i) = b(j)]$$

is the probability that the label is unchanged after τ steps. Values near one mean that the packaged distinction survives the timescale on which the layer operates. Because only equality of labels matters, $S_\tau(b; \mu) = S_\tau(1 - b; \mu)$. We use S_τ both for the stored input bits of the gate laboratories and for scoring candidate partitions in the discovery pipeline.

Route mismatch. Route mismatch (RM) asks whether microstates with the same label also have the same coarse future. With $R_\tau(i, \cdot)$ the coarse transition profile of Section 2, choose nonnegative weights w_i on microstates with positive total, and write $W_x = \sum_{i: f(i)=x} w_i$ for the weight of fiber x and $W = \sum_i w_i$ for the total. For each fiber with $W_x > 0$, the induced macro row is the weighted average profile

$$K_{\tau,f}(x, \cdot) = \sum_{i: f(i)=x} \frac{w_i}{W_x} R_\tau(i, \cdot),$$

and route mismatch is the weighted average distance of individual profiles from it:

$$\text{RM}_\tau(f) = \sum_{x: W_x > 0} \frac{W_x}{W} \sum_{i: f(i)=x} \frac{w_i}{W_x} \|R_\tau(i, \cdot) - K_{\tau,f}(x, \cdot)\|_1.$$

RM lies between 0 and 2. It is zero exactly when all positively weighted microstates in each fiber share the same coarse profile; when every weight is positive, as in all our experiments, $\text{RM}_\tau(f) = 0$ is equivalent to lumpability at horizon τ . RM is an average, not a worst case. We use uniform weights in the parity laboratory, and the stationary distribution of P in the phase diagram, the reversible-versus-erased comparison, and the discovery pipeline; the supplementary parity-versus-AND sweep of Appendix A uses uniform weights [9, 11, 12].

Closure defect. The closure defect compares two routes from a distribution to its coarse future. With Q_f and U_f as in Section 2,

$$\Delta_{\tau,f}(\mu) = \|Q_f(\mu P^\tau) - Q_f(U_f(Q_f(\mu))P^\tau)\|_1.$$

The first route evolves the micro-distribution and then reduces it. The second reduces first, replaces the micro-distribution by the packaged one, evolves, and reduces again. A large defect means that microstructure hidden by the lens still influences the coarse future [8, 10, 13]. A single probe μ can, however, give a zero defect by cancellation or because it is already packaged. In the noiseless CNOT laboratory at $\tau = 1$, for example, the uniform distribution gives $\Delta = 0$ for the output-bit lens, whose route mismatch is 1. The worst case over all probes has a simple form.

Lemma 1 (Worst-case closure defect). *Let ρ_x be the prototype of fiber x , and let $\bar{R}_\tau(x, \cdot) = \sum_j \rho_x(j) R_\tau(j, \cdot)$ be the prototype-averaged profile. Then*

$$\Delta_{\tau,f}^{\max} := \sup_{\mu} \Delta_{\tau,f}(\mu) = \max_{i \in Z} \|R_\tau(i, \cdot) - \bar{R}_\tau(f(i), \cdot)\|_1,$$

where the supremum runs over probability distributions on Z . In particular, $\Delta_{\tau,f}^{\max} = 0$ if and only if f is lumpable at horizon τ .

Proof. Since $Q_f(\mu P^\tau) = \sum_i \mu_i R_\tau(i, \cdot)$ and $U_f(Q_f(\mu)) = \sum_x (\sum_{i: f(i)=x} \mu_i) \rho_x$, we have $\Delta_{\tau,f}(\mu) = \|\sum_i \mu_i [R_\tau(i, \cdot) - \bar{R}_\tau(f(i), \cdot)]\|_1$. This is a convex function of μ , so its maximum over the probability simplex is attained at a point mass, which gives the formula. The maximum is zero if and only if every profile equals the prototype average of its own fiber; then all profiles in a fiber coincide, which is lumpability. Conversely, if the profiles in each fiber coincide, their prototype average equals the common profile. $\square$

Together, S_τ , RM, and $\Delta_{\tau,f}$ make the first two clauses of Definition 1 measurable: stability measures persistence, and RM and the closure defect measure whether closure has been achieved rather than assumed.

3.2 Gate-channel and information diagnostics

Gate behavior is measured through induced input–output channels. Let f_{in} label the input at the start of the window and f_{out} label the output at the end. For each input label u we start from the uniform distribution on the microstates with $f_{\text{in}} = u$, evolve for τ steps, and read the output label y . This defines a channel $C_\tau(y | u)$. All gate metrics are computed from this channel with a uniform prior over input patterns.

If $t(u)$ is the intended truth-table output for input u , the *truth error* is

$$\text{err}_{\text{truth}} = \frac{1}{|U|} \sum_{u \in U} [1 - C_\tau(t(u) | u)].$$

To measure how deterministic the channel is without reference to an intended table, the *induced error*

$$\text{err}_{\text{induced}} = \frac{1}{|U|} \sum_{u \in U} \left(1 - \max_y C_\tau(y | u)\right)$$

grades the channel against its own best-fitting deterministic table. The conditional entropy

$$H(\text{out} | \text{in}) = -\frac{1}{|U|} \sum_{u \in U} \sum_y C_\tau(y | u) \log_2 C_\tau(y | u),$$

with $0 \log 0 = 0$, measures the same uncertainty without reducing each row to its largest entry.

For the accounting comparison we use the mutual information $I(\text{in}; \text{out}) = H(\text{out}) - H(\text{out} \mid \text{in})$ under the same uniform input prior, and define the *unretained input information*

$$U_{\text{loss}} = H(\text{in}) - I(\text{in}; \text{out}) = H(\text{in} \mid \text{out}),$$

the uncertainty about the input that remains after the output has been read. We also report the entropy drop $\Delta H = H(\text{in}) - H(\text{out})$, but the main comparison uses U_{loss} : an output can carry extra entropy that has nothing to do with the input, so a small or even negative entropy drop does not show that input information has been retained. This comparison is framed by the literature on irreversibility, reversible computation, and the thermodynamics of information [1, 2, 4, 19]. All Shannon quantities are in bits.

3.3 Entropy-production audit

To connect information processing with irreversibility we compute stationary entropy production. For a stationary distribution π of P , let $J_{ij} = \pi_i P_{ij}$ be the probability flux from i to j . The entropy-production rate is the relative entropy between forward and reversed fluxes,

$$\sigma(P) = \sum_{i,j} J_{ij} \log \frac{J_{ij}}{J_{ji}} = \sum_{i < j} (J_{ij} - J_{ji})(\log J_{ij} - \log J_{ji}),$$

in nats per step. Each pair contributes a nonnegative amount; a pair with flux in only one direction contributes $+\infty$, and a pair with no flux contributes zero. For a lens f and horizon τ , the *apparent* entropy production $\sigma_{\text{app}}(f, \tau)$ applies the same formula to the coarse fluxes $\sum_{i: f(i)=x} \sum_{j: f(j)=y} \pi_i (P^\tau)_{ij}$, computed from the pushforward of the same stationary law π rather than from a separately solved macro law. It is measured in nats per window of τ steps; dividing by τ gives nats per micro step. Because coarse-graining a pair of distributions cannot increase their relative entropy, $\sigma_{\text{app}}(f, \tau)$ never exceeds $\sigma(P^\tau)$. It is a Markov-approximation audit of the coarse description, not a claim that the coarse process is Markovian [20, 21]. In this paper entropy production is an audit variable, not a stand-alone thermodynamic claim.

3.4 Finite logic laboratories

Each gate laboratory has a register space X of bit strings and a degeneracy factor d , so that $Z = X \times \{1, \dots, d\}$. One tick of the dynamics applies a deterministic register update, flips the output bit with probability p_{gate} , flips each stored input bit independently with probability p_{mem} , and finally spreads the probability uniformly over the d microstates of the resulting register value. The last step makes packaging explicit: all microstates of one register value have identical transition rows, so the full register lens is exactly lumpable by construction. All experiments use $d = 3$.

There are three gate laboratories. The NOT laboratory has registers (a, c) , with stored input a and update $c' = 1 - a$. The AND laboratory has registers (a, b, c) , with stored inputs a, b and update $c' = a \wedge b$. The CNOT laboratory has registers (a, b) , with stored control a and target update $b' = a \oplus b$. The stored inputs are memory: the update leaves them unchanged, and only noise can flip them. Memory noise is set directly or through a staging barrier, $p_{\text{mem}} = 0.05 e^{-\text{barrier}}$. The phase-diagram sweep covers $p_{\text{gate}} \in \{0, 0.02, 0.05, 0.1\}$, $\text{barrier} \in \{0, 2, 6\}$, and $\tau \in \{1, 2, 4\}$ for all three gates, giving 108 conditions. The output lens reads c for NOT and AND and the target b for CNOT; the input lens reads a for NOT, (a, b) for AND, and the full initial register (a, b) for CNOT.

The fourth laboratory is not a gate. The parity-sector laboratory has two bits (a, b) and a degeneracy factor $d = 3$. From each register value it moves, with probability $1 - p_{\text{leak}}$, to one of the two values of the same parity chosen uniformly, and with probability p_{leak} it leaks to the other parity sector. The leak is biased: from the even sector it reaches 01 with probability $0.8 p_{\text{leak}}$ and 10 with probability $0.2 p_{\text{leak}}$; from the odd sector it reaches 11 with probability $0.7 p_{\text{leak}}$ and 00 with probability $0.3 p_{\text{leak}}$. The total leak out of each sector is the same, but the bias makes the future profiles of the individual bits differ between the two parity sectors, which produces nonzero route mismatch for lenses that hide parity. This laboratory asks which binary coarse variables stay closed as leakage and horizon vary, over $p_{\text{leak}} \in \{0.02, 0.05, 0.1, 0.2\}$ and $\tau \in \{1, 2, 4\}$.

3.5 Discovery pipeline

The discovery pipeline asks whether stable bits and gates can be recovered from the kernel without declaring them. For input bits we follow spectral approaches to metastable decomposition [16, 22]. Let D be the diagonal matrix of the stationary distribution, which is positive in every laboratory we analyze. The reversibilized operator

$$S = \frac{1}{2}(D^{1/2}P^\tau D^{-1/2} + (D^{1/2}P^\tau D^{-1/2})^\top)$$

is symmetric. We take an eigenvector v for its second-largest eigenvalue and return to micro coordinates through $\varphi = D^{-1/2}v$. Then φ is an eigenfunction of the additive reversibilization $A_\tau = \frac{1}{2}(P^\tau + D^{-1}(P^\tau)^\top D) = D^{-1/2}SD^{1/2}$, so the cut acts on a function of microstates rather than on the conjugated vector; φ need not be an eigenfunction of P^τ itself. Coordinates of φ that agree to numerical precision are treated as equal, and a candidate binary partition is formed at each gap between distinct values. Each candidate b is scored by $S_\tau(b; \pi) - \text{RM}_\tau(b)$, with stationary weights, so the preferred cut is both persistent and closed. When the second eigenvalue is repeated, the eigenvector is not unique; the pipeline is a heuristic that proposes good partitions, not an enumeration of all of them. Partition recovery is evaluated in the NOT laboratory (barrier 8, $p_{\text{gate}} = 0.01$) and the parity-sector laboratory ($p_{\text{leak}} = 0.05$), both at $\tau = 1$.

Output bits are found separately, in a NOT laboratory with barrier 6, $p_{\text{gate}} = 0.05$, and $\tau = 1$. States are first grouped into behavioral classes whose outgoing and incoming transition probabilities agree after rounding, which is a finite-precision heuristic rather than an exact equivalence. Binary partitions of these classes, other than the input partition and its complement, are then ranked by information gain $\Delta I_{\text{out}} = I_{\text{future}} - I_{\text{current}}$, where I_{future} is the mutual information between the discovered input bit now and the candidate output bit τ steps later, and I_{current} is the mutual information between the two at the same time [10, 23, 24]. A good output bit predicts the future without restating the present input. A candidate is accepted only if its gain is strictly positive, so the identity kernel and a completely mixing kernel, which we use as negative controls, yield no output bit. Finally, we compute the exact induced channel between the discovered partitions and, separately, fit a one-input gate to a balanced sample of 20000 transitions. Truth values need a convention: each discovered partition labels the class containing microstate 0 as 0. Reversing the labels of exactly one partition exchanges NOT and COPY without changing any information or error measure.

In summary, each clause of Definition 1 is tied to a measurement: stability to S_τ , closure to RM and the closure defect, gate semantics to channel error and entropy, and auditability to unretained input information and entropy production. The next four sections read these diagnostics off the frozen bundle to ask which coarse variables are robust, which gates close under stress, what changes under erasure, and whether bits and gates can be discovered rather than imposed.

4 Parity Robustness

The parity-sector laboratory is built around leakage between the even-parity states $\{00, 11\}$ and the odd-parity states $\{01, 10\}$ of a two-bit register, so the natural candidate coarse variable is the parity bit $a \oplus b$. We compare it with two kinds of alternatives on the same four register values: the familiar Boolean predicate $a \wedge b$, and balanced binary partitions, which split the four values into two pairs. The comparison is about closure rather than prediction: the measured quantity is route mismatch, so smaller is better.

Up to swapping the two labels there are only three balanced partitions: parity, the first bit a , and the second bit b . Each appears twice among the six labelings that assign two values to each label. The frozen experiment draws 60 such labelings at random with replacement and reports their median RM, so the “random” baseline in fact samples these three partitions. Because the family is this small, the comparison can be made exactly.

Proposition 1 (Exact route mismatch in the parity-sector laboratory). *Let $p = p_{\text{leak}} \in [0, 1]$, let $\tau \geq 1$, and use uniform weights. Then*

$$\begin{aligned} \text{RM}_\tau(a \oplus b) &= 0, & \text{RM}_\tau(a) &= \frac{p}{2} |1 - 2p|^{\tau-1}, & \text{RM}_\tau(b) &= \frac{p}{10} |1 - 2p|^{\tau-1}, \\ \text{RM}_\tau(a \wedge b) &= \frac{2}{3} \left| \frac{1}{2} - \frac{6p}{5} \right| |1 - 2p|^{\tau-1}, & S_\tau(a \oplus b; \mu) &= \frac{1 + (1 - 2p)^\tau}{2} \text{ for every } \mu. \end{aligned}$$

Consequently, for $0 < p < \frac{1}{2}$ the median RM over the six balanced labelings equals $\text{RM}_\tau(b) > 0$, whereas the parity lens has RM exactly zero at every horizon.

Proof. Every microstate’s transition row depends only on the parity of its register value: rows of the two even values coincide, as do rows of the two odd values, and all d copies of a value share its row. Write r_E and r_O for the even and odd rows. The same holds for P^τ , whose rows are $r_E P^{\tau-1}$ and $r_O P^{\tau-1}$. Hence each parity fiber contains a single profile and $\text{RM}_\tau(a \oplus b) = 0$. Since the parity flips with probability p at each step independently of everything else, it is unchanged after τ steps with probability $[1 + (1 - 2p)^\tau]/2$, whatever the starting state. For the other lenses, let $\delta = r_E - r_O$. The total mass of δ on the even sector is $(1 - p) - p = 1 - 2p$ and on the odd sector is $-(1 - 2p)$; because P has row r_E on even states and r_O on odd states, $\delta P = (1 - 2p)\delta$, and so the even and odd rows of P^τ differ by $(1 - 2p)^{\tau-1}\delta$. The lenses a and b put one even and one odd value in each fiber, so at horizon τ each profile lies at distance $\frac{1}{2}|1 - 2p|^{\tau-1}$ times the ℓ_1 norm of the pushed-forward δ from the fiber average. Pushing δ forward gives $(\frac{p}{2}, -\frac{p}{2})$ through a and $(-\frac{p}{10}, \frac{p}{10})$ through b , which yields the stated values. For $a \wedge b$, the fiber $\{11\}$ is a single value and contributes nothing, while the fiber $\{00, 01, 10\}$, which carries three quarters of the weight, contains one even and two odd values; the even and odd profiles differ by $(1 - 2p)^{\tau-1}(\frac{1}{2} - \frac{6p}{5})$ in their mass on 11, and averaging the resulting distances gives the stated formula. Among the six balanced labelings the RM values are $\text{RM}_\tau(a \oplus b)$, $\text{RM}_\tau(a)$, and $\text{RM}_\tau(b)$, each twice; since $0 < \text{RM}_\tau(b) < \text{RM}_\tau(a)$ for $0 < p < \frac{1}{2}$, the median is $\text{RM}_\tau(b)$. $\square$

Figure 2 compares the frozen experimental values with these closed forms. The parity lens beats the sampled median at all 12 grid points (p_{leak}, τ) . Its frozen RM values are all below 1.4×10^{-16} , which is floating-point residue of an exact zero; an independent recomputation of all twelve points in rational arithmetic gives exactly 0 for parity and exactly $p_{\text{leak}}|1 - 2p_{\text{leak}}|^{\tau-1}/10$ for the median, for example $1/200$ at $p_{\text{leak}} = 0.05$, $\tau = 1$ and $1/50$ at $p_{\text{leak}} = 0.20$, $\tau = 1$. The sampled median coincides with the exhaustive median at every grid point. The AND lens is far from closed on the same slices, with RM 0.293 and 0.173.

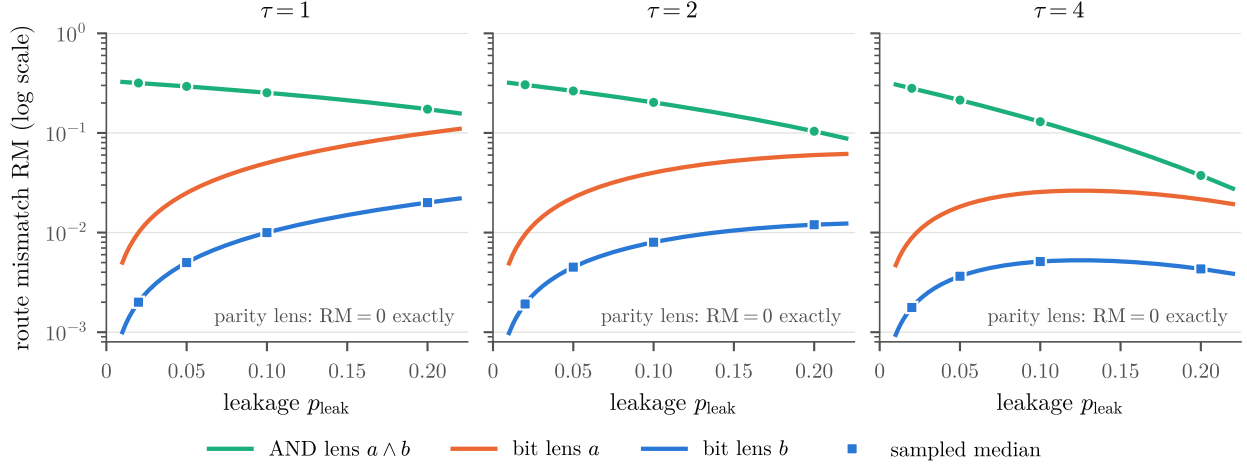


Figure 2. Route mismatch in the parity-sector laboratory, on a logarithmic scale. Curves are the closed forms of Proposition 1; markers are the frozen experimental values for the AND lens (circles) and for the median of 60 sampled balanced labelings (squares), which coincides with the bit lens b . The parity lens is not drawn because its RM is exactly zero at every leakage rate and horizon.

The result is useful because parity stays closed even as it becomes a noisy signal. By Proposition 1, the probability that the parity bit has changed after τ steps is $[1 - (1 - 2p_{\text{leak}})^\tau]/2$, which grows from 0.05 at $p_{\text{leak}} = 0.05$, $\tau = 1$ to 0.4352 at $p_{\text{leak}} = 0.20$, $\tau = 4$, while its RM stays exactly zero. This is the distinction the framework is designed to expose. A coarse variable can become less predictable and still be the right closure variable for the layer: robustness here means that the packaged distinction continues to define a macro-state whose future does not depend on the hidden detail, not that its value is transmitted without error. Figure 2 also shows that the other partitions improve at long horizons only because their mismatch decays like $|1 - 2p_{\text{leak}}|^{\tau-1}$ as the chain mixes, not because they become closed.

The contrast with AND is the structural point. AND is not an adversarial choice; it is a familiar Boolean predicate on the same bits. In this laboratory, however, it cuts across the sector structure instead of following it, and its RM exceeds the median balanced partition at every grid point. Semantically familiar predicates are not automatically closure-friendly. A substrate does not inherit its logic from our preferred vocabulary; its dynamics determines which distinctions it can stabilize.

This is the sense of XOR in the title. In this laboratory parity is the packaged distinction that the dynamics itself respects, by construction. That is a designed demonstration, not a theorem that parity dominates other variables in arbitrary substrates. The general lesson is more modest: variables aligned with a system’s dominant sector or invariant structure can be exactly closed coarse variables when other natural candidates are not [8–10]. The next section asks how far closure extends from a single robust variable to complete gates under noise, staging, and timescale stress.

5 Gate Closure Phase Diagrams

We now move from a single robust variable to complete gates under stress. Logic in these laboratories is not simply present or absent; closure occupies regimes. The frozen phase bundle covers 108 conditions: three gates, three staging barriers, three evaluation horizons τ , and four output-noise levels p_{gate} . For each condition it records the truth error relative to the intended one-step gate, the

conditional entropy $H(\text{out} \mid \text{in})$, and the route mismatch of the output-bit lens, weighted by the stationary distribution. Figure 3 shows all three diagnostics for every condition.

The channels have closed forms, which explains the patterns in the figure. Write $m = p_{\text{mem}}$, $g = p_{\text{gate}}$, and let $q_s = [1 - (1 - 2m)^s]/2$ be the probability that a stored bit has flipped an odd number of times in s ticks. At the last tick the output is computed from the stored inputs as they were one tick earlier, so for $\tau \geq 1$ the NOT truth error is

$$\text{err}_{\text{truth}}^{\text{NOT}} = g + q_{\tau-1} - 2g q_{\tau-1},$$

and for AND the probability that the output is 1, given initial inputs (a, b) , is $g + (1 - 2g) r_a r_b$ with $r_0 = q_{\tau-1}$ and $r_1 = 1 - q_{\tau-1}$. For CNOT, τ ideal applications act as the identity when τ is even and as CNOT when τ is odd. The control enters the target an odd number of times only when τ is odd, and exactly $\lfloor \tau/2 \rfloor$ independent memory flips of the control survive in the target, so the target differs from the τ -step ideal map with probability

$$e_\tau = \frac{1}{2} [1 - (1 - 2g)^\tau (1 - 2m)^{\lfloor \tau/2 \rfloor}],$$

independently of the input. The truth error against the one-step XOR table is therefore e_τ for odd τ and $\frac{1}{2}$ for even τ . An independent audit recomputes every metric in all 108 conditions from these formulas and recomputes every RM value from rational macro-kernels and explicitly constructed stationary laws, without importing the pipeline’s stationary solver or channel code (Appendix B). All conditions agree, and truth error, induced error, and conditional entropy rise monotonically with output noise throughout the grid.

Averaged over barriers and horizons, the NOT truth error rises from 0.0236 at $p_{\text{gate}} = 0$ to 0.1189 at $p_{\text{gate}} = 0.1$, and the AND truth error from 0.0225 to 0.1180. The CNOT average rises only from 0.3333 to 0.3667, because two thirds of its conditions have even τ , where the error is $\frac{1}{2}$ by construction. Averages hide most of the structure: barrier and horizon matter about as much as output noise, and the three diagnostics do not move together in the same way for every gate.

NOT and AND: closure fails when storage drifts. With strong staging and a short window, both gates are recovered cleanly. At barrier 6, $\tau = 1$, and $p_{\text{gate}} = 0$, NOT and AND both have $\text{err}_{\text{truth}} = 0$, $H(\text{out} \mid \text{in}) = 0$, and output-lens RM = 0.000496. The small RM is not zero for a structural reason. The output bit alone does not determine its own future, which depends on the stored inputs; in our Lean formalization, the noise-free NOT update provably fails to descend to the output bit on the whole state space. Stationary weighting, however, concentrates almost all mass on states whose output already agrees with the gate, so the average mismatch is small. Under weak staging and longer windows the picture changes even without output noise, because the stored inputs themselves drift. At barrier 0 ($p_{\text{mem}} = 0.05$), $\tau = 4$, and $p_{\text{gate}} = 0$, NOT reaches truth error 0.1355 = q_3 , entropy 0.572, and RM 0.139, and AND reaches truth error 0.126, entropy 0.497, and RM 0.142. The symbolic gate has not changed; the carrier has become unreliable, and with it the closure.

CNOT: a perfect channel on an unclosed bit. CNOT behaves differently because its output lens reads the target bit alone. At $\tau = 1$ and $p_{\text{gate}} = 0$, CNOT is an exact gate channel at every barrier: $\text{err}_{\text{truth}} = 0$ and $H(\text{out} \mid \text{in}) = 0$. Yet the output-lens RM is 1, the largest value RM can take for a two-label lens: two microstates with the same target bit but different controls have opposite futures. At $\tau = 1$ with output noise, the RM is exactly $|1 - 2p_{\text{gate}}|$ at every barrier. This is not a contradiction. A perfect input–output channel says that the full input determines the output; closure says that the output label determines its own future. The Lean formalization proves that

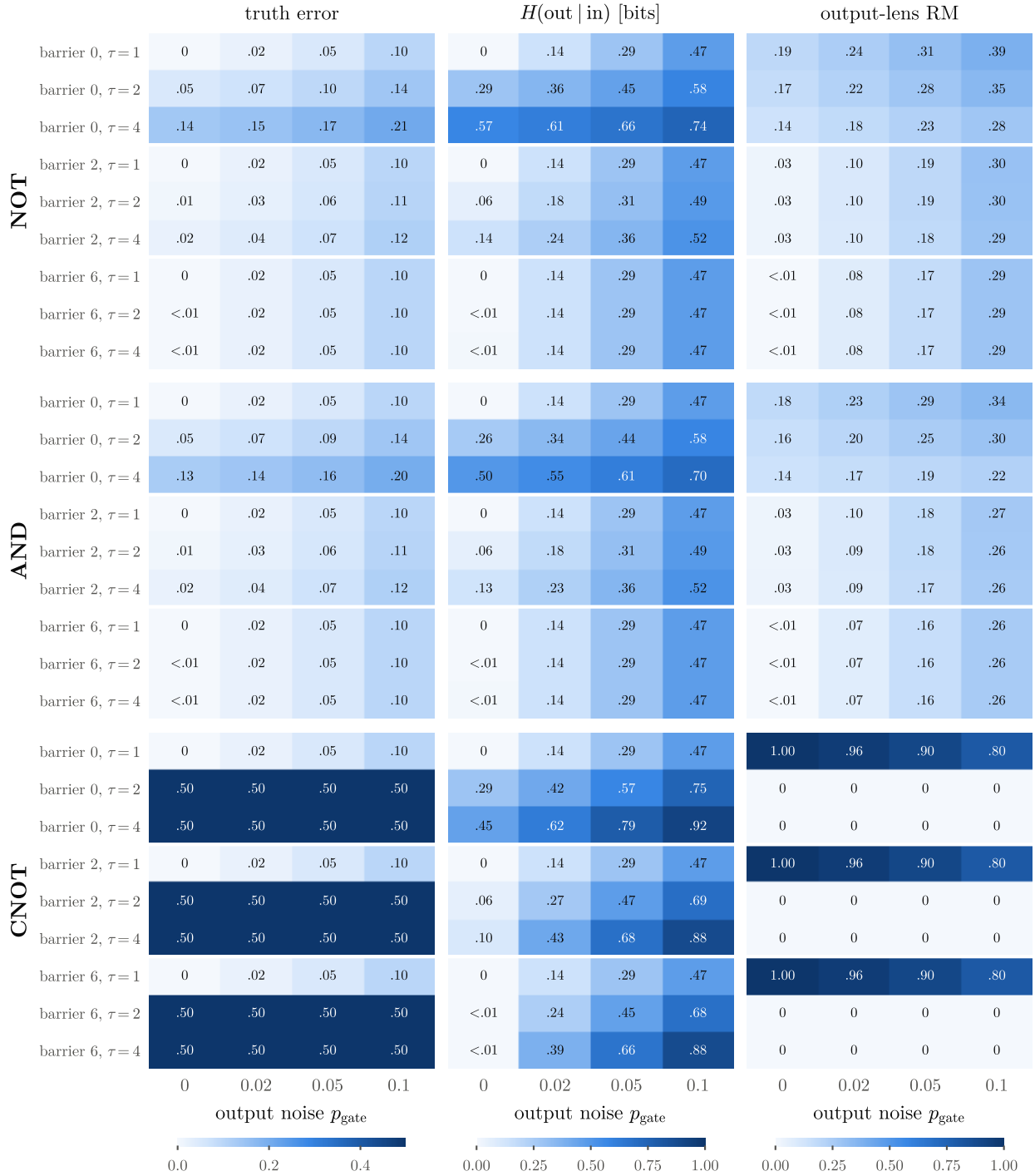


Figure 3. Gate-closure phase portrait for all 108 conditions. Rows within each block are staging barrier and evaluation horizon τ ; columns are output noise p_{gate} . Columns of panels show truth error against the one-step truth table, conditional entropy, and stationary-weighted route mismatch of the output-bit lens; each column of panels shares one color scale, and cell labels give the values. Entries below 10^{-12} are shown as 0. For CNOT at even τ the truth error is 0.5 by construction, because two ideal applications return the target to its initial value.

Table 1. Three views of one CNOT laboratory ($p_{\text{gate}} = 0.02$, $p_{\text{mem}} = 0$, $\tau = 1$, $d = 3$). The input is the two-bit register in every row, so $H_{\text{in}} = 2$ bits. Δ is the closure defect for a point-mass probe on a microstate with register value $(1, 0)$; for the erased output this probe attains the worst case of Lemma 1. RM uses stationary weights. EPR is the stationary entropy-production rate of the micro-kernel for the micro readout and the apparent rate for the two coarse views, in nats per step. Entries shown as 0 are exactly zero; the frozen floating-point values are below 10^{-15} .

View	States	Closure and audit			Channel information [bits]				
		Δ	RM	EPR	H_{out}	$H(\text{out} \mid \text{in})$	I	ΔH	U_{loss}
Micro readout	12	0	0	0	3.585	1.726	1.859	-1.585	0.141
Retained register	4	0	0	0	2	0.141	1.859	0	0.141
Erased output	2	0.96	0.96	0	1	0.141	0.859	1	1.141

the one-step CNOT update admits no induced map on the target bit alone. The reversible two-bit gate is closed on its full register, while the one-bit output lens is not.

At even horizons the roles reverse. Two ideal applications return the target to its initial value, so the one-step XOR table is the wrong reference and the truth error is $\frac{1}{2}$ by construction. The induced error, which grades the channel against its own best table, stays small at barrier 6 and $p_{\text{gate}} = 0$: 0.000124 at $\tau = 2$ and 0.000248 at $\tau = 4$, with conditional entropy 0.0018 and 0.0033. The output-lens RM is exactly 0 at every even horizon, barrier, and noise level, because e_τ does not depend on the input: after an even number of ticks the target is its initial value plus independent noise, and so it does determine its own future. Lean proves the noise-free version of this statement, that the two-tick CNOT update descends to the target bit. The $\tau > 1$ rows of CNOT are therefore closure stress tests, not repeated grading of one truth table: the relevant macro-operator changes with the horizon.

Read as a whole, Figure 3 is a phase portrait of embodied logic. Output noise raises channel error in the expected way, but staging and horizon decide whether that noise is the dominant failure or one contributor among several. For NOT and AND the central transition runs from a short-horizon, high-barrier regime in which the gate is closed in practice to a long-horizon, weak-staging regime in which memory drift drives both entropy and mismatch up. For CNOT the lesson is that closure depends on retaining the right carrier: a reversible gate can be channel-perfect at one horizon while its one-bit output lens is maximally unclosed. The next section makes this explicit by comparing the retained and erased views of one CNOT laboratory.

6 Reversible Embedding versus Erased XOR

Table 1 isolates the accounting point of the paper. All three rows describe the same CNOT laboratory with the same parameters: degeneracy $d = 3$, output noise $p_{\text{gate}} = 0.02$, no memory noise, and a single tick, $\tau = 1$. The substrate does not change between rows; only the description presented to the layer does. In every row the input is the two-bit register (a, b) , so $H_{\text{in}} = 2$ bits. The *micro readout* reads the output as the full microstate, one of 12. The *retained register* reads it as the four-valued register (a, b) . The *erased output* reads only the target bit, which carries $a \oplus b$, and forgets the control. In Six Birds terms the split happens at packaging (P5): keeping or discarding side information changes the carrier of the layer, and accounting (P6) records the consequences.

The retained register is closed and loses little. Since all microstates of a register value share a transition row, the retained register is exactly lumpable: its closure defect and route mismatch

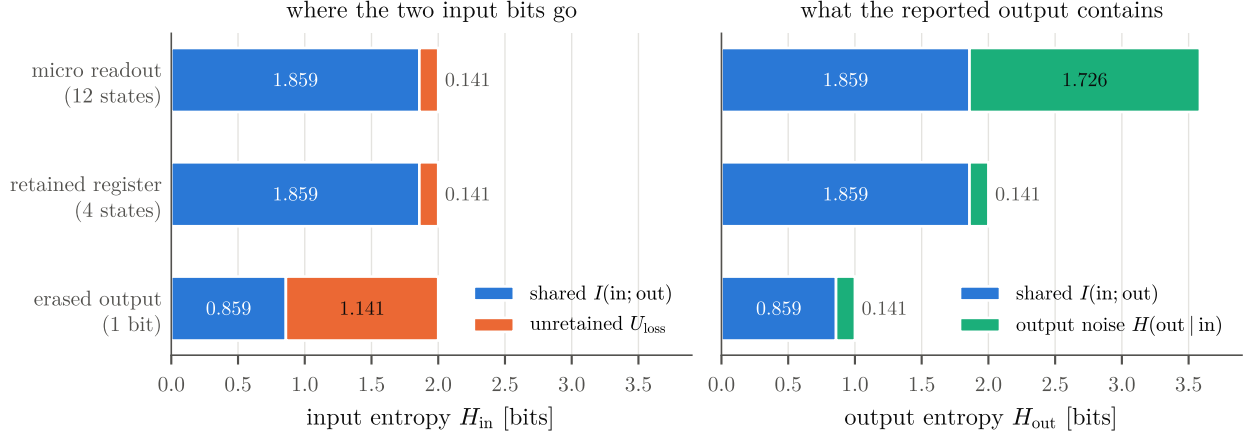


Figure 4. Information budget of the three views in Table 1. Left: the two input bits split into information shared with the output, $I(\text{in}; \text{out})$, and unretained input information, $U_{\text{loss}} = H(\text{in} | \text{out})$. Right: the output entropy splits into the same shared information and output noise, $H(\text{out} | \text{in})$. The micro readout and the retained register share the same information; the micro readout only adds $\log_2 3$ bits of degeneracy noise. Erasing the control removes exactly one bit of shared information and raises U_{loss} from $h_2(0.02)$ to $1 + h_2(0.02)$.

are zero. Its channel is a noisy copy of a permutation of the four register values, so, writing $h = h_2(0.02) \approx 0.1414$ for the binary entropy of the output noise, $H_{\text{out}} = 2$, $H(\text{out} | \text{in}) = h$, $I(\text{in}; \text{out}) = 2 - h \approx 1.859$, and $U_{\text{loss}} = h \approx 0.141$. The only input information the output fails to account for is the bit flipped by noise.

The micro readout adds entropy but no information. Reading the output at full microscopic resolution changes the output entropies but not the information about the input. The destination microstate is drawn uniformly among the three copies of the register value, independently of everything else, so it adds $\log_2 3 \approx 1.585$ bits to both H_{out} and $H(\text{out} | \text{in})$, which become 3.585 and 1.726. The mutual information and U_{loss} are unchanged, and the entropy drop becomes negative, $\Delta H = -\log_2 3$. This row illustrates why we use U_{loss} rather than entropy drop: a richer output can have more entropy than the input without carrying any more information about it.

The erased output is a different object. Forgetting the control changes both closure and accounting. The route mismatch of the erased output is exactly $24/25 = 1 - 2p_{\text{gate}} = 0.96$: microstates with the same target bit but different controls have nearly opposite futures, exactly as in the phase diagram of Section 5. The closure defect of the point probe is also 0.96, and by Lemma 1 no probe gives a larger value, so the probe does not hide a larger closure failure. On the information side, the output entropy drops to one bit, the conditional entropy stays at h because the noise on the reported bit is unchanged, and so $I(\text{in}; \text{out}) = 1 - h \approx 0.859$ and $U_{\text{loss}} = 1 + h \approx 1.141$. The ratio of unretained input information between the erased and retained views is

$$\frac{1 + h}{h} = 8.0701 \dots$$

That the ratio exceeds eight is equivalent to $h_2(0.02) < 1/7$, which we certify with rational upper and lower bounds on the logarithms involved (Appendix B). Figure 4 shows where the input information goes in each view.

Table 2. Unsupervised recovery of the stable bit in the NOT laboratory (barrier 8, $p_{\text{gate}} = 0.01$) and the parity-sector laboratory ($p_{\text{leak}} = 0.05$), both at $\tau = 1$. Agreement is the fraction of microstates on which the top-ranked cut matches the intended bit, up to relabeling. Candidates are the distinct threshold cuts of the spectral eigenfunction. The score is $S_\tau - \text{RM}$ with stationary weights; the RM of both winners is exactly zero, with frozen floating-point values below 10^{-16} .

Laboratory	Agreement	Candidates	Top-ranked cut			Runner-up scores	
			S_τ	RM	Score	2nd	3rd
NOT	1	3	1.0000	0	1.0000	0.970	0.970
Parity-sector	1	3	0.9500	0	0.9500	0.432	0.429

Two further features of Table 1 are instructive. First, the conditional entropy is the same, h , in the retained and erased views. The noise on the reported bit has not changed; what has changed is how much of the input the reported channel can still carry once the control is forgotten. Second, all three entropy-production rates are exactly zero. With no memory noise the control never changes, so the chain is reducible; on the uniform stationary law that we use for it, the kernel is symmetric and therefore in detailed balance. The decisive audit signal is thus not a coarse thermodynamic signature but the joint change in closure and retained information. Reading the laboratory through the erased lens does not physically erase a register, and it does not by itself establish a Landauer heat cost.

This is the central packaging-and-accounting lesson. Packaging is not an innocent matter of notation. The retained register keeps the full reversible carrier and supports a closed two-bit operator. The erased output reports a one-bit summary and turns a reversible gate into a many-to-one channel with large closure failure. What looks like “the same XOR gate viewed more coarsely” is, as an audited layer, a different object with different closure properties and an eightfold larger loss ledger. Discussions of physical logic often treat an irreversible description as a reversible one with some bookkeeping omitted. The comparison shows that omitting the bookkeeping changes the closure status of the layer itself. The next section turns from declared views to the inverse problem: recovering stable bits and a gate from transition structure alone.

7 Discovering Bits and Gates

The last result reverses the usual direction. So far we have declared candidate bits and gate lenses and asked whether they close. Here we ask whether they can be recovered from the transition kernel without being named in advance. The discovery pipeline of Section 3 starts from the kernel alone: it proposes binary partitions from spectral structure, scores them by stability minus route mismatch, and only then interprets the winners as packaged predicates. For outputs, it ranks partitions by how much they predict about the future beyond what they restate about the present.

Stable bits. Table 2 shows that partition recovery is exact in both laboratories: the top-ranked cut agrees with the intended bit on every microstate. Each laboratory yields three candidate cuts. Because all copies of a register value share a transition row and a stationary weight, an eigenfunction of the reversibilization A_τ for a nonzero eigenvalue, as here, is constant on each register value, takes at most four distinct values, and so admits at most three threshold cuts. In the NOT laboratory the winning cut is the stored input bit, with stability 0.99998, which is one minus the memory-flip probability $0.05e^{-8}$, and RM zero; both runners-up score 0.970. In the parity-sector laboratory the

Table 3. Recovered NOT gate in the NOT laboratory (barrier 6, $p_{\text{gate}} = 0.05$, $\tau = 1$). The exact column is computed from the kernel between the discovered input and output partitions; the sample column fits a one-input gate to 20000 transitions drawn equally from the two input classes, with starting microstates uniform within each class. Truth bits list the output for inputs 0 and 1 under the convention that each partition labels the class containing microstate 0 as 0.

Quantity	Exact induced channel	Balanced sample
Truth bits (declared orientation)	[1, 0]	[1, 0]
Gate error	0.050	0.04845
$H(\text{out} \mid \text{in})$ [bits]	0.2864	0.2798
Transitions (input 0 $\rightarrow$ 0/1; input 1 $\rightarrow$ 0/1)	–	488/9512; 9519/481
Input partition score $S_\tau - \text{RM}$		0.99988
Output partition $I_{\text{future}} / I_{\text{current}}$ [bits]		0.7136 / 0

winner is the parity bit, with stability $0.95 = 1 - p_{\text{leak}}$ and RM zero, while the runners-up fall to 0.432 and 0.429. The margin is large in the parity laboratory and narrow but clear in the NOT laboratory. Since the eigenvector need not be unique when eigenvalues repeat, the pipeline is a heuristic for proposing good partitions, not a guarantee that it finds every good partition or every latent bit.

A NOT gate. Once the input partition has been found, the harder question is which binary partition of behavioral classes is the output. Table 3 records the answer. The discovered input partition has score 0.99988. The selected output partition has $I_{\text{future}} = 0.7136$ bits and $I_{\text{current}} = 0$: it says nothing about the present input, yet after one tick it carries $1 - h_2(0.05)$ bits about it, the capacity of a binary symmetric channel that flips its output with probability 0.05. The exact induced channel between the two partitions has error 0.05, equal to p_{gate} , and conditional entropy $h_2(0.05) \approx 0.2864$ bits. A balanced sample of 20000 transitions gives the estimates 0.04845 and 0.2798, and its fitted truth bits [1, 0], or as a flattened one-step table [0, 1, 1, 0], are those of NOT.

One caveat matters for interpretation. Input and output partitions are discovered separately, and each has two possible orientations. Under the stated convention the recovered truth bits are [1, 0]; reversing exactly one orientation gives [0, 1], a COPY gate, with identical information and error. The pipeline therefore recovers the carriers and a nonconstant binary channel between them, and it identifies the gate as NOT in declared coordinates. It does not discover an intrinsic orientation shared by two otherwise unrelated truth alphabets. Even so, the gate is found only after the carriers on which it closes have themselves been discovered; it is not a relabeling of hand-specified registers.

Tables 2 and 3 complete the empirical arc of the paper. The earlier sections argued that logic should be diagnosed by stability, closure, and audit rather than assumed. The discovery results show that the same diagnostics also work in reverse: they first detect which packaged distinctions a substrate supports and then recover the operator induced on them. The framework does not merely restate known bits and gates in Six Birds vocabulary; in these laboratories a logic layer can be found from the dynamics alone.

8 Discussion

We began with a question that comes before the usual implementation story: when does a substrate earn a logic layer? The answer developed through Definition 1 asks for more than labels 0 and 1. It

asks for packaged predicates that persist, induced operators that close on them, and an audit that measures mismatch and loss. In this sense logic is not primitive; it is earned by closure. The four results each sharpen a different part of that answer.

Parity is a measurement, not a mystique. In the parity-sector laboratory the parity lens has route mismatch exactly zero at every leakage rate and horizon, whereas the other two balanced partitions and the AND predicate have strictly positive mismatch at every tested point, with explicit closed forms (Proposition 1). XOR does not occupy a privileged metaphysical tier. Parity is the distinction that this laboratory’s sector structure respects, and the diagnostics detect that. A substrate can make some logical forms cheap and exact without making every familiar Boolean predicate equally natural. To “XOR a stone” is to find that parity can be the first logical distinction a substrate stabilizes.

Gate closure is a regime, not a verdict. NOT and AND close in short-horizon, high-barrier regimes and degrade over longer horizons under weaker staging, even without output noise, because their stored inputs drift. CNOT shows a different failure: a noiseless one-step CNOT is a perfect input–output channel while its one-bit output lens is maximally unclosed, and at even horizons the same lens becomes exactly closed again. A perfect input–output channel is not the same thing as a closed state description. Different gates fail in different ways because closure depends on the carrier the layer retains, not only on noise.

Forgetting side information changes the logical object. The retained and erased views of one CNOT laboratory differ in route mismatch and worst-case closure defect by 0.96 and in unretained input information by a factor of 8.07. That is not well described as “the same gate with a larger bill.” It changes what counts as the state of the layer. Packaging and accounting are inseparable here: packaging fixes the carrier, and the audit shows that discarding the control turns a closed reversible operator into a many-to-one report with very different closure properties [4, 19–21]. The micro readout makes a complementary point: describing the output more finely adds entropy but no information about the input, so entropy drop alone does not measure what a description has lost.

The framework runs in both directions. The diagnostics that test closure also discover it. The pipeline recovers the intended stable bits exactly in both benchmark laboratories, and it reconstructs a NOT gate whose exact channel error equals the laboratory’s output noise. Because each partition’s orientation is a convention, the recovered gate is NOT in declared coordinates; what is discovered without convention is the pair of carriers and the nonconstant channel between them. The pipeline starts from transition structure, not from declared variables, so it offers a way to ask from the dynamics alone which distinctions deserve logical status and which operators act on them.

Scope and limitations. The laboratories are finite Markov chains chosen for diagnostic clarity, not for physical realism, and the results are demonstrations within them rather than claims about natural substrates. Definition 1 is an operational criterion with application-dependent tolerances; we report the diagnostics instead of a single binary verdict on whether a system “has” a logic layer, and specifying thresholds would be necessary for such a verdict. We do not implement an independent solver for feasibility constraints. Route mismatch is an average over a chosen weighting, and a single-probe closure defect can vanish by cancellation; we therefore pair the probe with the worst-case defect where it matters. Apparent entropy production is computed per evaluation window

and audits a Markov approximation; it is not a measurement of dissipated heat. The discovery pipeline is limited to binary partitions and one-input gates and relies on finite-precision heuristics for behavioral classes and repeated eigenvalues. Finally, in the set-based laboratories studied here the expressible events always form a Boolean algebra, so the paper does not exhibit non-Boolean event semantics; it measures which Boolean operations a layer can hold stable and execute. For partition-based routes from classical state spaces to complementary observables and non-Boolean propositional structure, see [7, 18]; for the quantum-logical background, see [17]; and for a Six Birds extension in that direction, see [25].

Outlook. The broader consequence is pluralistic. The question is not whether logic is “really there” behind every coarse description, but which closures a layer can sustain, at which horizon, on which carrier, and at what audit cost. A layer may hold a parity bit exactly while conjunction leaks, may support a reversible operator only on a larger carrier than the one being read, or may express a complement it cannot execute. The Six Birds viewpoint does not eliminate the diversity of logics; it ties each to the closures a substrate can actually sustain. If XOR teaches anything, it is not that parity is magical but that substrates tell us which distinctions deserve logical status. Logic is not what a theorist can name at a layer; it is what a substrate can stabilize, close, and audit there.

A Reproducibility and artifacts

Reproduction. The repository regenerates every experimental output used in the paper. The command `make reproduce-all` reruns the six experiment families: the parity-robustness experiment, the gate phase diagram, the reversible-versus-erased comparison, the partition-discovery run, the NOT gate-discovery run, and a supplementary parity-versus-AND sweep over CNOT and AND laboratories that is not discussed in the main text. The command `make seal` additionally runs the test suite, freezes the headline claims, validates the frozen state, and runs the independent mathematical audit and the Lean axiom audit described in Appendix B. The generated files are listed in `results/final_manifest.json` with their hashes and sizes.

Frozen claims. The paper is written against a frozen bundle under `results/final_claims/`. The file `claims.json` holds the headline scalars, and CSV files hold the data behind each figure and table. Figures and tables are rendered from this bundle, and from the gate-discovery source record for the exact channel values in Table 3, by `scripts/render_paper_assets.py`. A validator checks the hashes and sizes of all source artifacts, rebuilds the entire frozen numerical interface from them in isolation, applies the independent mathematical audit, and checks the headline results directly: the parity win at every grid point, exact partition recovery, and the eightfold loss ratio. The test suite confirms that the validator rejects both a modified source artifact and a forged claim.

Code and data availability. All code for the laboratories, diagnostics, discovery pipeline, frozen claims, and paper assets is in the public repository <https://github.com/ioannist/six-birds-logic>. Core implementations are under `src/emergent_logic/`, experiment entry points under `experiments/`, configurations under `configs/`, and reproduction and validation scripts under `scripts/`. The Lean formalization is under `lean/LogicClosure/`, and the mathematical review that accompanies the code is `docs/mathematical_review.txt`. No external or proprietary data are used. Shannon quantities are in bits; entropy-production rates are in nats per step, or per evaluation window of τ steps for apparent rates.

B Formal and exact support

The paper’s claims rest on three kinds of support: theorems proved in Lean, closed-form results checked in exact rational arithmetic, and floating-point computation. This appendix states which claims rest on which.

B.1 Lean formalization

The Lean 4 development (toolchain 4.28.0) formalizes the exact predicate and quotient core of Section 2. It concerns sets, predicates, and deterministic updates; it does not formalize the stochastic laboratories, the floating-point experiments, or any inference from definability to metastability. Table 4 summarizes its contents. An audit script builds the project and inspects the transitive axioms of all 32 named theorems, including a trivial build check. Only Lean’s standard foundations `propext`, `Quot.sound`, and `Classical.choice` occur; there are no local axioms and no `sorry` or `admit`. Classical choice enters through the Boolean-valued factorization, whose extension of a macro-predicate off the image of the lens uses classical decidability; it is not an algorithm for arbitrary infinite spaces.

Table 4. Contents of the Lean formalization.

Module	What is proved
Definability	A predicate factors through a lens if and only if it is constant on fibers, with no surjectivity or inhabitation assumptions, in both proposition-valued and Boolean-valued forms; definable predicates contain true and false and are closed under conjunction, disjunction, negation, and exclusive or; definability is preserved by pullback along an update that descends to the labels.
Quotient dynamics	An update descends to a quotient if and only if it respects the equivalence relation; the induced map is unique and compatible with composition.
Finite parity example	The parity relation on $\text{Fin } 4$ is exactly the kernel of the parity lens $\text{Fin } 4 \rightarrow \text{Fin } 2$; the lens is onto; collapsing each state to a parity representative preserves the lens and induces the identity update on the quotient.
Gate examples	The noise-free NOT, AND, and CNOT register updates descend to their stored inputs; CNOT is an involution and injective; the one-step CNOT update admits no induced map on the target bit alone, while two ticks do descend to it; the NOT output bit alone does not descend on the whole state space.

B.2 Exact numerical certificates

An independent audit script re-derives the main experimental quantities without importing the pipeline’s stationary solver or gate-channel code, and compares them with the frozen values.

Parity laboratory. For all twelve grid points, exact rational arithmetic enumerates the six balanced labelings and evaluates RM for each. It confirms that parity has RM exactly 0, that the exhaustive median equals the closed form of Proposition 1, and that the sampled median exceeds the parity value at every point. The supplementary parity-versus-AND sweep is checked in the same way at all 24 of its conditions, including worst-case closure defects for the full register and parity lenses.

Phase diagram. Every metric in all 108 conditions is compared with the channel formulas of Section 5, and every RM value is recomputed from rational macro-kernels and explicitly constructed stationary laws. The grid is checked for completeness and for duplicate conditions. The memory noise $0.05 e^{-\text{barrier}}$ is irrational at the nonzero grid barriers; the exact checks use each parameter’s recorded decimal value and compare with the floating-point outputs to a stated tolerance.

Reversible versus erased. The erased output has RM exactly $24/25$. The loss ratio $(1 + h)/h$ with $h = h_2(1/50)$ exceeds eight if and only if $h < 1/7$. For $1 \leq q \leq 2$, let $t = (q - 1)/(q + 1)$; then

$$\ln q = 2 \sum_{k \geq 0} \frac{t^{2k+1}}{2k + 1},$$

and after N terms the remaining tail is nonnegative, strictly positive when $q > 1$, and at most $2t^{2N+1}/((2N + 1)(1 - t^2))$. Applying these bounds to $q = 2$, $32/25$, and $49/25$, and using $\ln(1/50) = \ln(32/25) - 6 \ln 2$ and $\ln(49/50) = \ln(49/25) - \ln 2$, encloses $h_2(1/50)$ in a rational interval that lies below $1/7$. The resulting rational enclosure of the ratio is $[8.07010862677, 8.070108626771]$.

What remains floating-point. Shannon quantities other than those above, the barrier exponentials, the spectral eigenvectors of the discovery pipeline, and all sampled statistics are floating-point computations. Values that are exactly zero in rational arithmetic appear in the frozen bundle as residues below 10^{-14} ; the paper reports them as zero and says so where they appear.

C Version history

Version 1 (March 9, 2026). Initial release.

Version 2 (March 9, 2026). Expanded citations to the related literature and consolidated the bibliography. The results were unchanged.

Version 3 (October 3, 2026). This version follows a mathematical review of the code, the Lean development, and the manuscript. The four main conclusions are unchanged. The substantive changes are:

1. The route-mismatch formula now states the normalization by fiber mass that the code always used, and its relation to lumpability on the positively weighted support. Reported RM values are unchanged.
2. The closure-defect discussion adds Lemma 1 on the worst case over all probes, and confirms that the probe used for the erased CNOT view attains it.
3. In the reversible-versus-erased comparison, the micro row previously used the four-state output lens and duplicated the retained register. It now reads the full microstate, which gives $H_{\text{out}} = 3.585$, $H(\text{out} \mid \text{in}) = 1.726$, and $\Delta H = -1.585$ bits. Its mutual and unretained input information, and the eightfold ratio, are unchanged.
4. The spectral discovery pipeline now cuts the eigenfunction $D^{-1/2}v$ in micro coordinates and groups numerically equal coordinates before forming thresholds. Both laboratories now yield three candidate cuts instead of five, and the runner-up scores changed; the winning partitions and their exact agreement are unchanged. The gate-discovery record adds the exact induced channel

and raw counts, and the label-orientation convention that distinguishes NOT from COPY is stated.

5. Entropy production is computed in pair-flux form, coarse rates use the pushforward of the same stationary law, and apparent rates are reported per evaluation window. Reported values are unchanged up to floating-point residue.
6. Closed forms for the parity laboratory (Proposition 1) and the gate channels were added, together with exact rational audits and a rational certificate for the eightfold ratio. The Lean development was extended from three claims to the converse and bridge theorems of Table 4.
7. The text distinguishes semantic complement from executable negation, describes $E_{\tau,f}$ as evolve, reduce, and lift, and states the operational status of Definition 1. The abstract and exposition were rewritten for readability, the parity and phase figures were redrawn, and Figure 4 is new.

References

- [1] R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.
- [2] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973. doi: 10.1147/rd.176.0525.
- [3] Edward Fredkin and Tommaso Toffoli. Conservative logic. *International Journal of Theoretical Physics*, 21(3-4):219–253, 1982. doi: 10.1007/BF01857727.
- [4] Charles H. Bennett. The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982. doi: 10.1007/BF02084158.
- [5] Ioannis Tsiokos. Six Birds: Foundations of Emergence Calculus, January 2026. URL <https://arxiv.org/abs/2602.00134>. arXiv:2602.00134 [cs.LO].
- [6] Zdzisław Pawlak. Rough sets. *International Journal of Computer & Information Sciences*, 11(5):341–356, 1982. doi: 10.1007/BF01001956.
- [7] David Ellerman. The logic of partitions: Introduction to the dual of the logic of subsets. *The Review of Symbolic Logic*, 3(2):287–350, 2010. doi: 10.1017/S1755020310000018.
- [8] Cosma Rohilla Shalizi and Cristopher Moore. What is a macrostate? subjective observations and objective dynamics. *Foundations of Physics*, 55:2, 2025. doi: 10.1007/s10701-024-00814-1. Article 2.
- [9] Herbert A. Simon and Albert Ando. Aggregation of variables in dynamic systems. *Econometrica*, 29(2):111–138, 1961. doi: 10.2307/1909285.
- [10] L. Barnett and A. K. Seth. Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108(1):014304, 2023. doi: 10.1103/PhysRevE.108.014304.
- [11] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer-Verlag, New York, 1976. ISBN 9780387901923.
- [12] Peter Buchholz. Exact and ordinary lumpability in finite markov chains. *Journal of Applied Probability*, 31(1):59–75, 1994. doi: 10.2307/3215235.
- [13] Dror Givon, Raz Kupferman, and Andrew Stuart. Extracting macroscopic dynamics: model problems and algorithms. *Nonlinearity*, 17(6):R55–R127, 2004. doi: 10.1088/0951-7715/17/6/R01.

- [14] Ioannis Tsiokos. To Count a Stone with Six Birds: A Mathematics is A Theory, January 2026. URL <https://zenodo.org/records/18402004>.
- [15] Ioannis Tsiokos. To Lay a Stone with Six Birds: Finite-State Semantics for Packaging, Directionality, and Coarse-Graining, February 2026. URL <https://www.preprints.org/manuscript/202602.1699>. Preprint, posted February 27, 2026.
- [16] Peter Deuffhard and Marcus Weber. Robust perron cluster analysis in conformation dynamics. *Linear Algebra and its Applications*, 398:161–184, 2005. doi: 10.1016/j.laa.2004.10.026.
- [17] Garrett Birkhoff and John von Neumann. The logic of quantum mechanics. *Annals of Mathematics*, 37(4):823–843, 1936. doi: 10.2307/1968621.
- [18] Harald Atmanspacher and Peter beim Graben. Complementary observables and non-boolean logic outside quantum physics, October 2015. URL <https://arxiv.org/abs/1510.03325>. arXiv:1510.03325 [quant-ph].
- [19] Juan M. R. Parrondo, Jordan M. Horowitz, and Takahiro Sagawa. Thermodynamics of information. *Nature Physics*, 11(2):131–139, 2015. doi: 10.1038/nphys3230.
- [20] Massimiliano Esposito. Stochastic thermodynamics under coarse graining. *Physical Review E*, 85(4):041125, 2012. doi: 10.1103/PhysRevE.85.041125.
- [21] Kyogo Kawaguchi and Yohei Nakayama. Fluctuation theorem for hidden entropy production. *Physical Review E*, 88(2):022147, 2013. doi: 10.1103/PhysRevE.88.022147.
- [22] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3-4):817–879, 2001. doi: 10.1023/A:1010388907793.
- [23] William Bialek, Ilya Nemenman, and Naftali Tishby. Predictability, complexity, and learning. *Neural Computation*, 13(11):2409–2463, 2001. doi: 10.1162/089976601753195969.
- [24] James P. Crutchfield and Karl Young. Inferring statistical complexity. *Physical Review Letters*, 63(2):105–108, 1989. doi: 10.1103/PhysRevLett.63.105.
- [25] Ioannis Tsiokos. A Six-Birds’ Eye View of Quantum Theory: Defining Objects by What We Can Distinguish, Not What We Imagine, February 2026. URL <https://zenodo.org/records/18608833>.